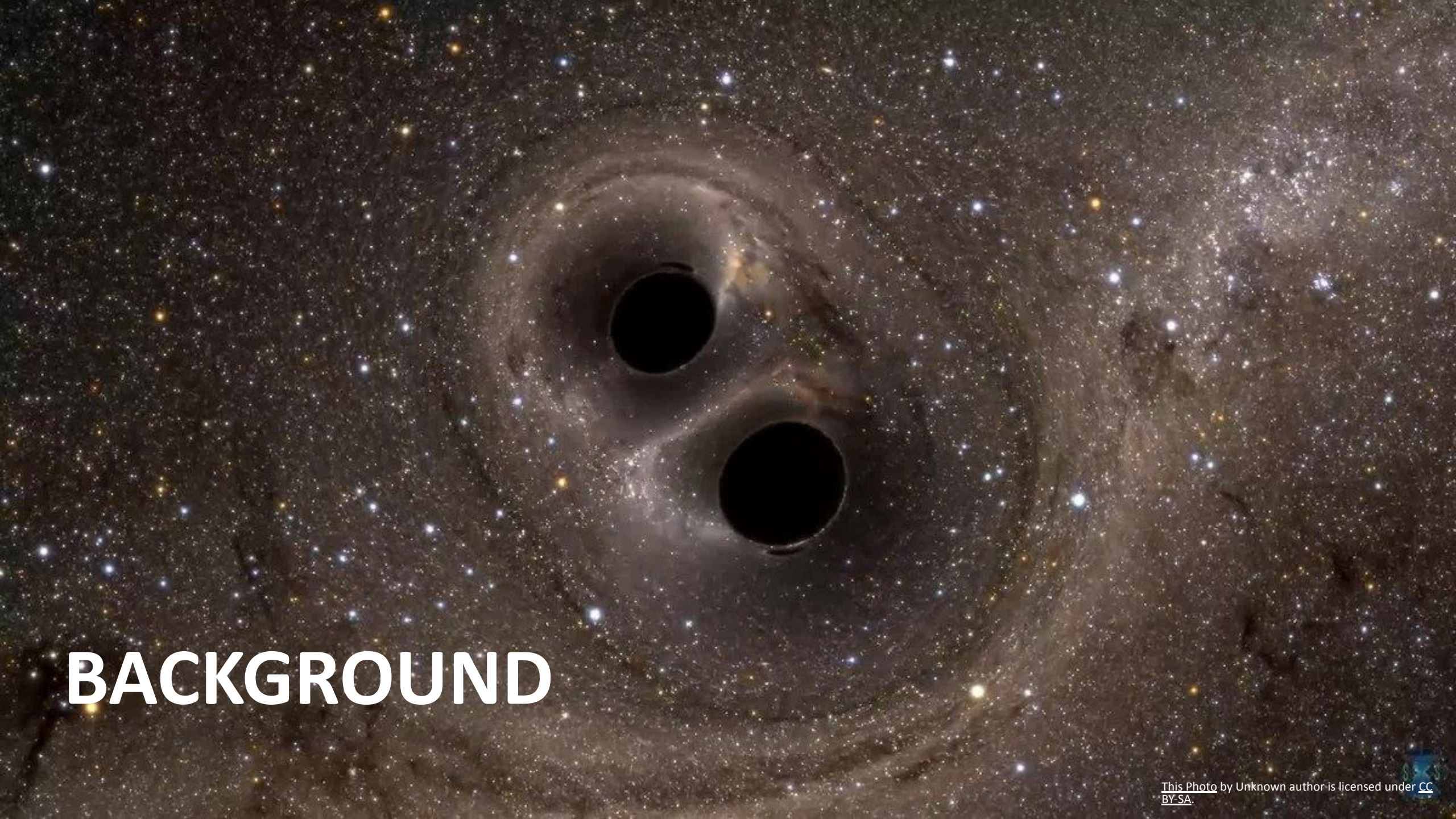


NEURAL ORDINARY DIFFERENTIAL EQUATIONS: LEARNING DYNAMICS FROM DATA

KEIGAN CULLEN

DR. SCOTT FIELD

DR. ALFA HERYUDONO



BACKGROUND

WHAT ARE GRAVITATIONAL WAVES?

- Predicted by Einstein with General Relativity
- Caused by accelerating massive bodies
- Analogous to a ball moving along the surface of water
 - Ball = Black hole
 - Water = Spacetime
 - Water waves = Gravitational Waves
- Binary black hole systems (BBH)

WHAT ARE GRAVITATIONAL WAVES?

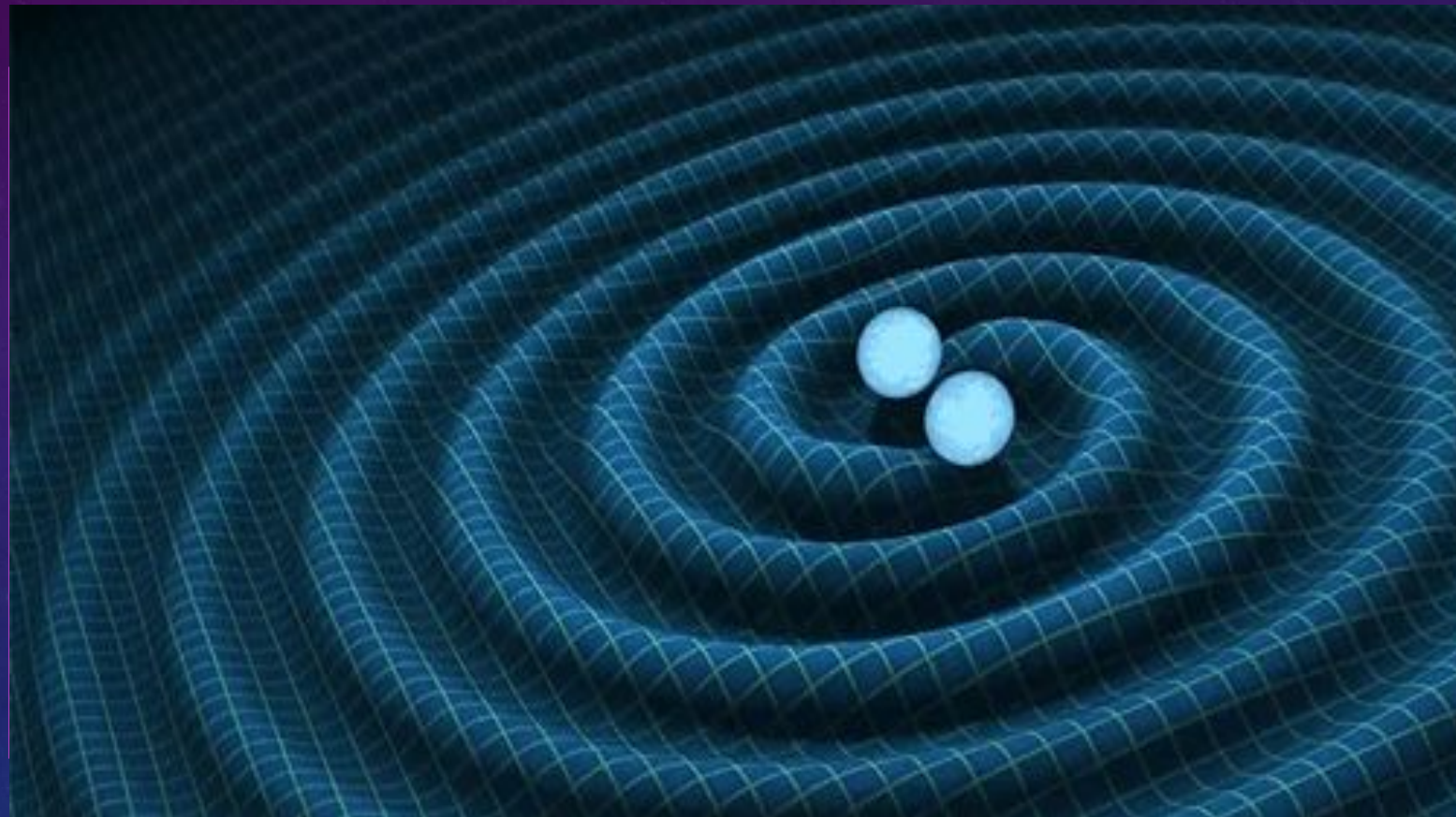


Fig 1: Animation of BBH. [This gif](#) is from user [b1-66er](#) on [gyfcat.com](#)

LIGO

- Laser Interferometer Gravitational-Wave Observatory
 - Louisiana, Washington (VIRGO, Italy)
- Basic ideas of interferometry circa 1960's
- First detection in September 2015
- Kip Thorne, Barry Barish, and Rai Weiss won the 2018 Nobel prize for founding LIGO

LIGO

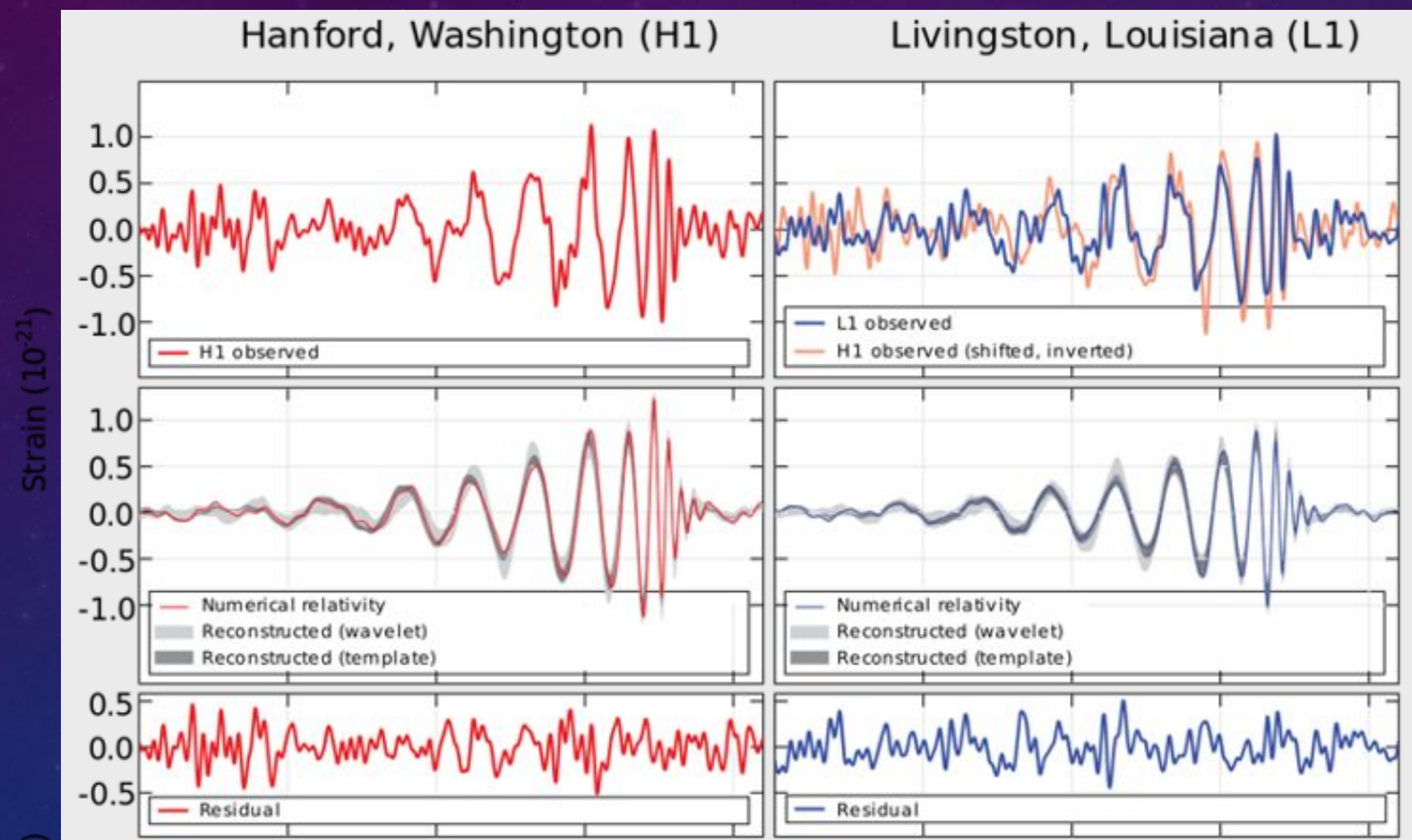


Fig 2: First observation of GWs. Observation of Gravitational Waves from a Binary Black Hole Merger.

WHAT ARE GWS USED FOR?

- Determining BBHs parameters
 - Mass, spin, location...
- Test for GR
- Potential probe for Quantum Gravity

WHY ARE GW MODELS IMPORTANT?

- Models predict shape of waveform
 - Real data is compared to model data
- If model disagreement > 0.01 , model is inaccurate:
 - Incorrect tests of GR
 - Incorrect parameters of BBH systems

GW MODELS

- Often described by system of ordinary differential equations
 - 2-Body Problem
 - Simple Harmonic Oscillator

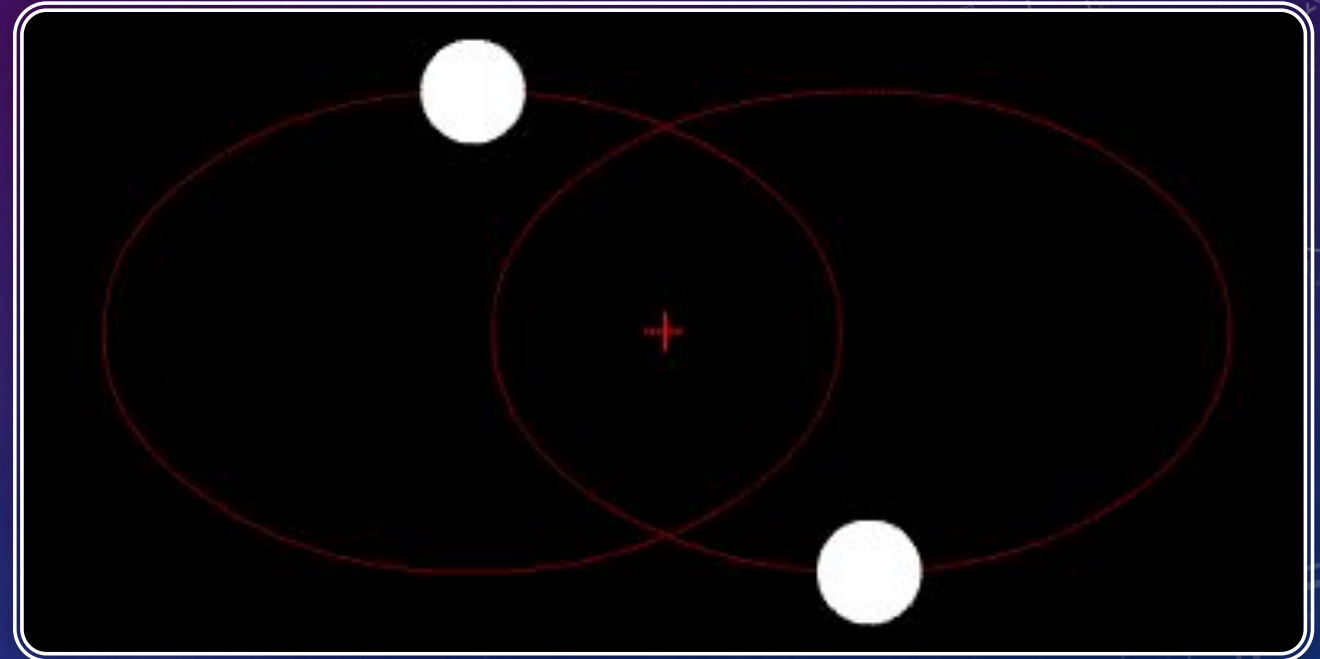


Fig 3: A simple 2-body system animation w/ fixed COM. [This Photo](#) by Unknown author is licensed under [CC BY-SA](#).

ODE EXAMPLE

Kepler's 3rd Law:

- Centripetal force = gravitational force

$$\begin{aligned}\frac{m_1 v^2}{r} &= G \frac{m_1 m_2}{r^2} \\ 4\pi^2 m_1 \frac{r}{T^2} &= G \frac{m_1 m_2}{r^2} \quad (v = 2\pi \frac{r}{T}) \\ \frac{r^3}{T^2} &= \frac{G m_2}{4\pi^2}\end{aligned}$$

$$\dot{r} = 0 \quad (\text{for circular orbit})$$

$$\dot{\theta} = \omega \quad (\text{angular frequency})$$

Fig 4: Kepler's 3rd law derivation (left). ODE for the system (right).
(created in Latex)

BBH MODELS

Non-relativistic

$$\Omega^2 = \frac{Gm}{r^3} \left\{ 1 + (-3 + \nu)\gamma + \left(6 + \frac{41}{4}\nu + \nu^2 \right) \gamma^2 + \left(-10 + \left[-\frac{75707}{840} + \frac{41}{64}\pi^2 + 22 \ln \left(\frac{r}{r'_0} \right) \right] \nu + \frac{19}{2}\nu^2 + \nu^3 \right) \gamma^3 \right\} + \mathcal{O} \left(\frac{1}{c^8} \right).$$

Fig 5: Relativistic correction of Kepler's 3rd Law. Image from [Luc Blanchet](#).

- "Same" 2-body problem
 - GR is required
- Post-Newtonian
 - Derived from Einstein's Equations
- Deriving these ODEs can be challenging

MOTIVATION

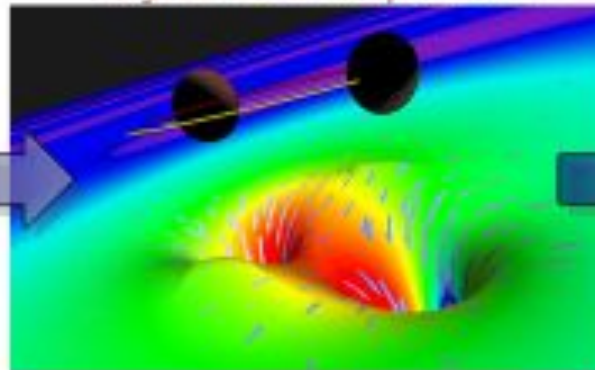
- Project goal is to directly discover the ODE of BBH
 - From LIGO signal or Post-Newtonian data
 - Desire is to be more computationally efficient

Instead of analyzing the physics to learn about GWs...

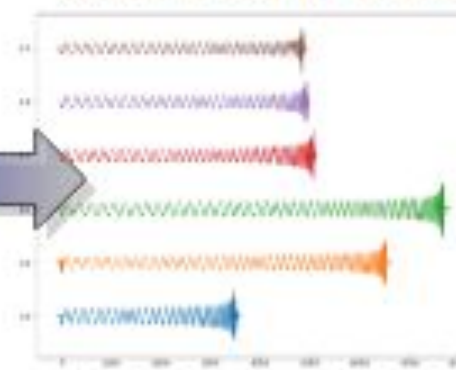
Einstein's equations



Dynamical equations

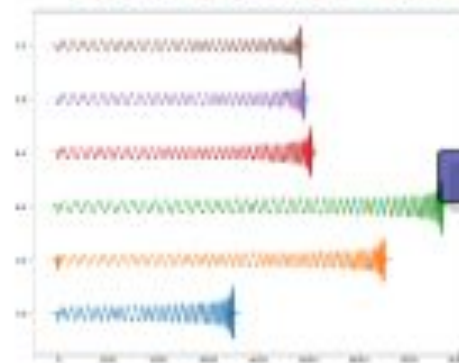


Gravitational waves



... we analyze GWs to learn about the physics

Gravitational waves



Dynamical equations

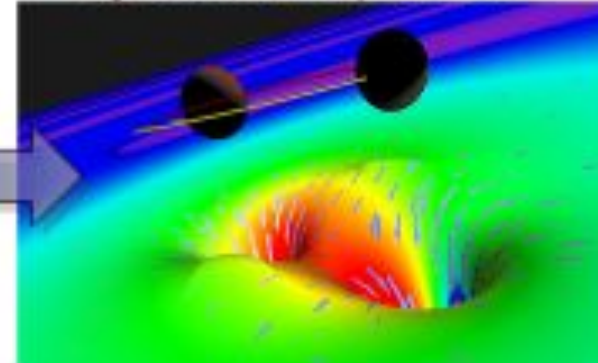


Fig 6: Driving motivation. Image from Brendan Keith.

SIMPLE HARMONIC OSCILLATOR

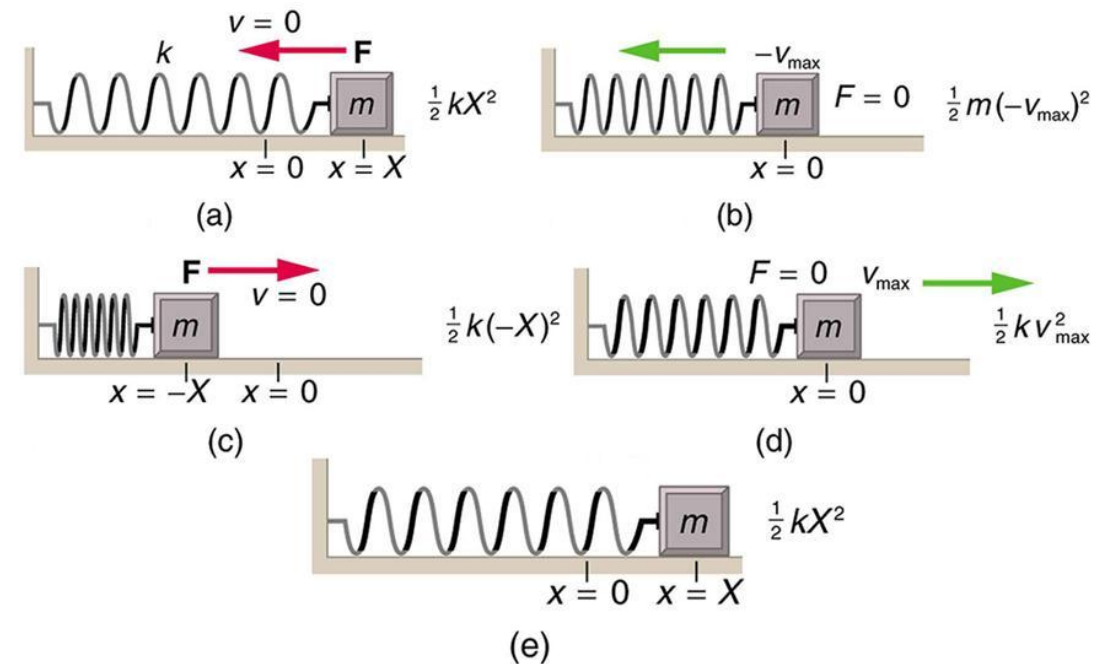
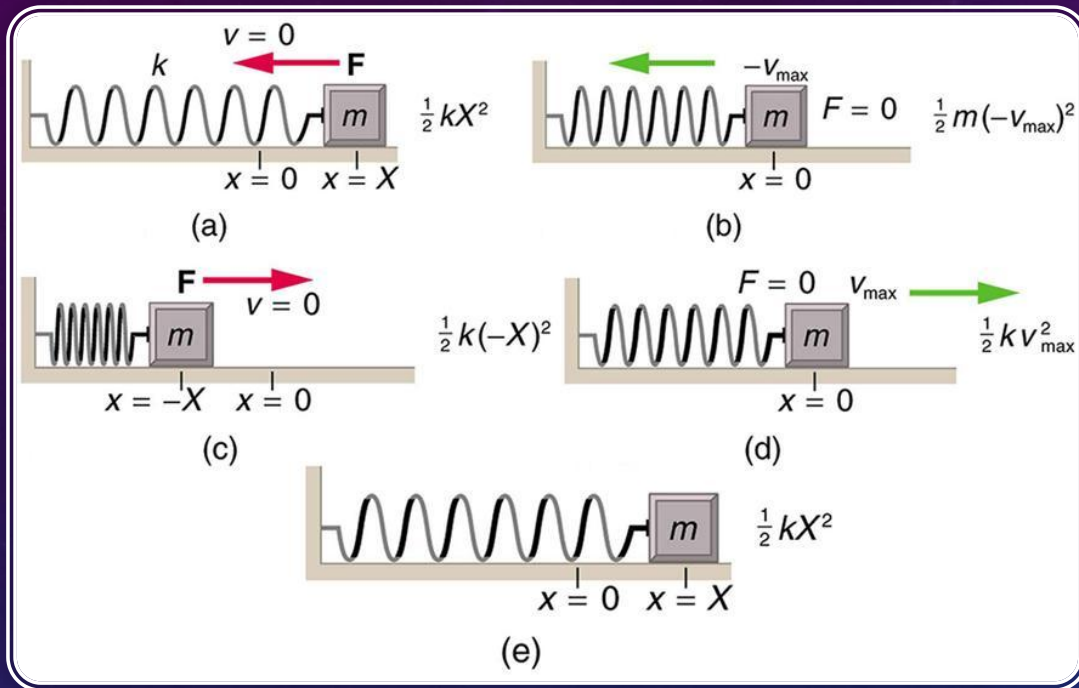


Fig 7: Oscillatory spring behavior.

THE SHO

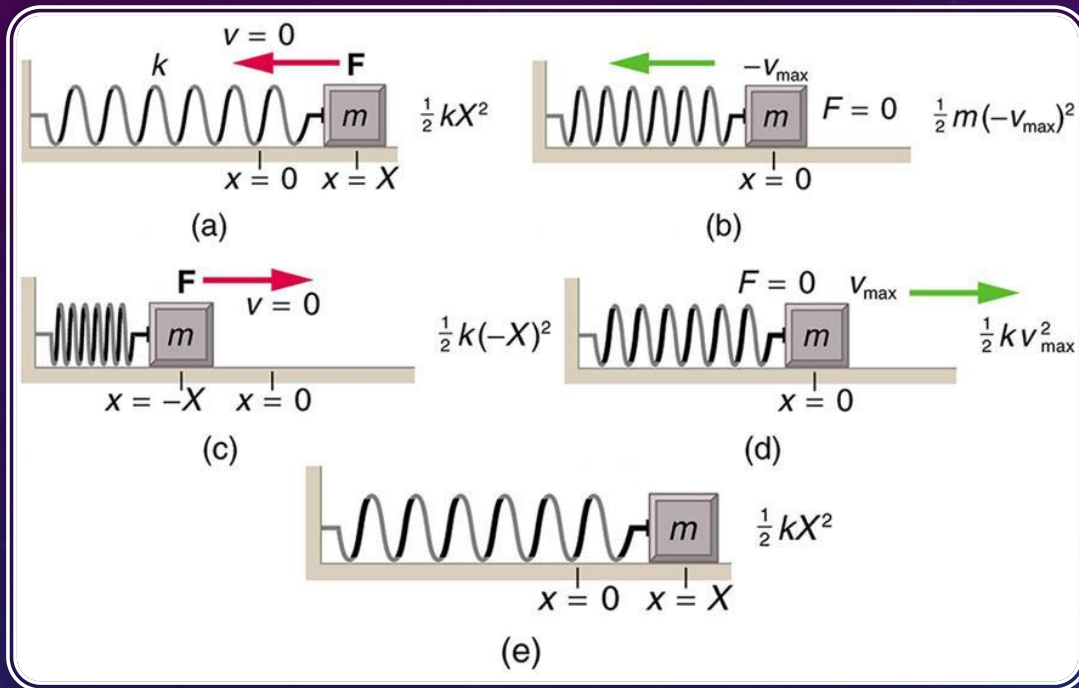


- Based on Hooke's Law ($F = -kx$)
- Apply Newton's Law ($F = ma$)
- Ordinary Differential Equation is found!:

$$-\frac{k}{m}x = \frac{d^2x}{dt^2}$$

Fig 8: Derivation of SHO ODE (Latex).

THE SHO



$m \equiv 1$ (simplicity)

$$\frac{d^2x}{dt^2} = -kx$$

$$\frac{dx}{dt} = p$$

$$\frac{dp}{dt} = -kx$$

Fig 9: Re-formulation of ODE. This is the form used in code (Latex).

PROBLEM SETUP

- "Hooke's law equations unknown"
 - Discover the ODE through machine learning
- Numerically simulate data
 - As opposed to collecting experimental data
- Exploring computational methods, packages, etc.
 - Code is built from scratch in Julia language
 - Apply similar methods to GW case

NEURAL ODE

- Feed-Forward Network
 - The inputs are multiplied by series of weights
 - Can compare outputs with desired values
 - Adjust weights through training
 - Ideal for machine learning

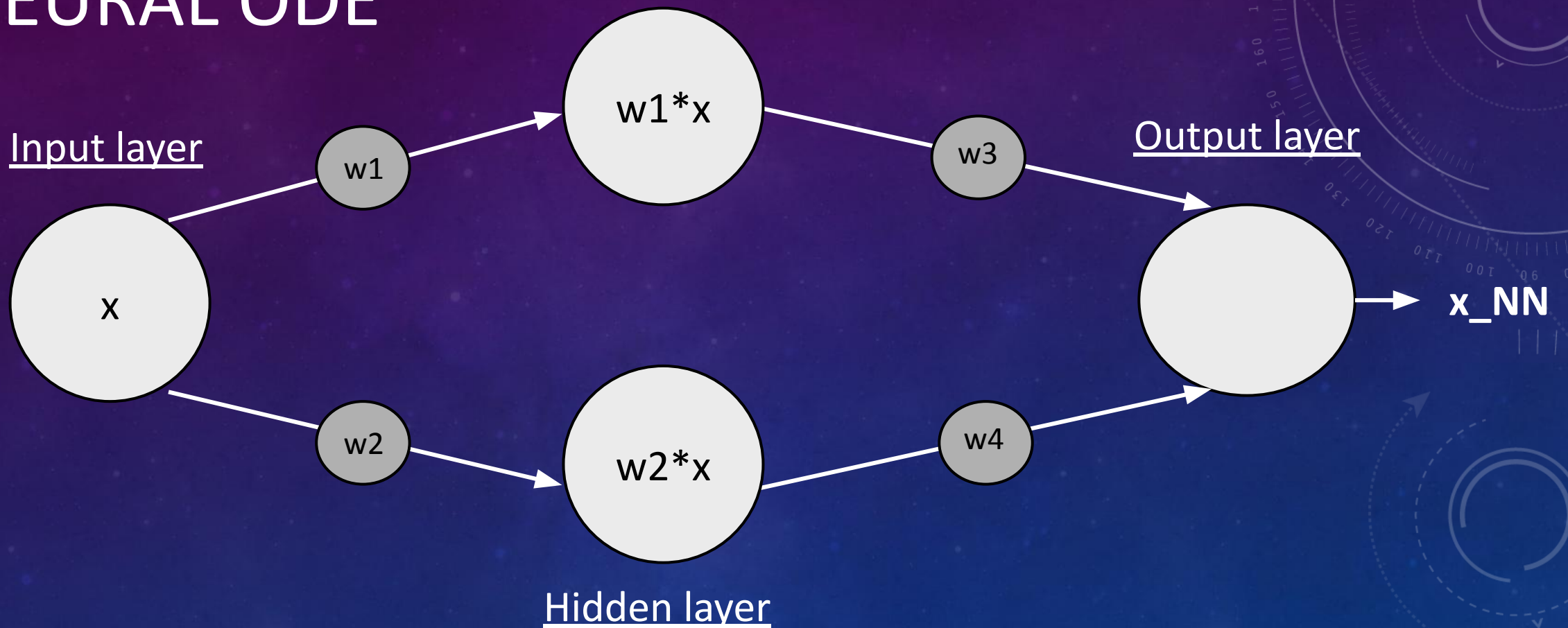
$$\frac{dx}{dt} = p$$

$$\frac{dp}{dt} = NN(x; \lambda) \quad (\lambda^* = \text{network weights/biases})$$

*this is what the network is optimizing for

Fig 10: The neural network in place of the ODE (Latex).

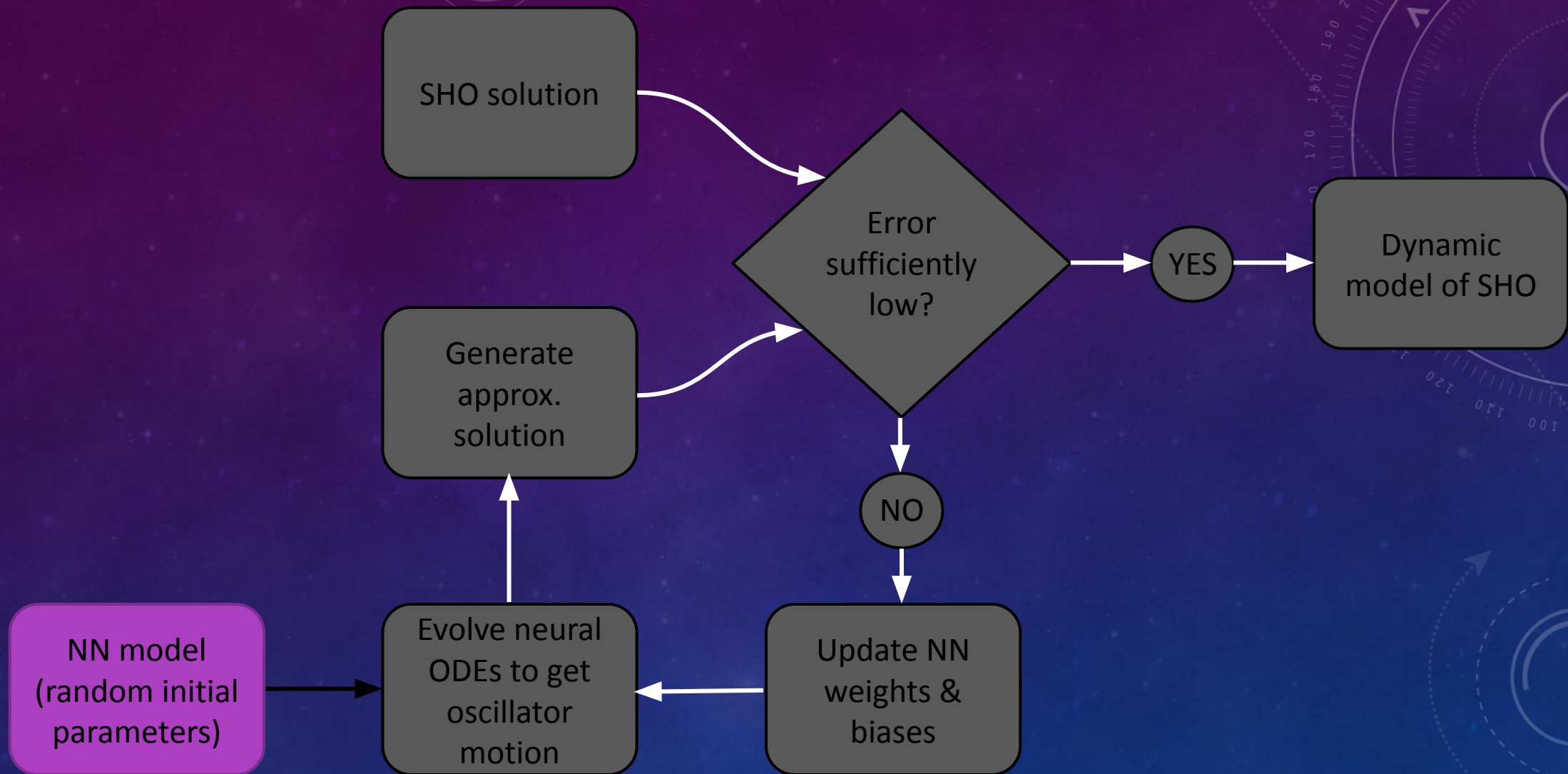
NEURAL ODE



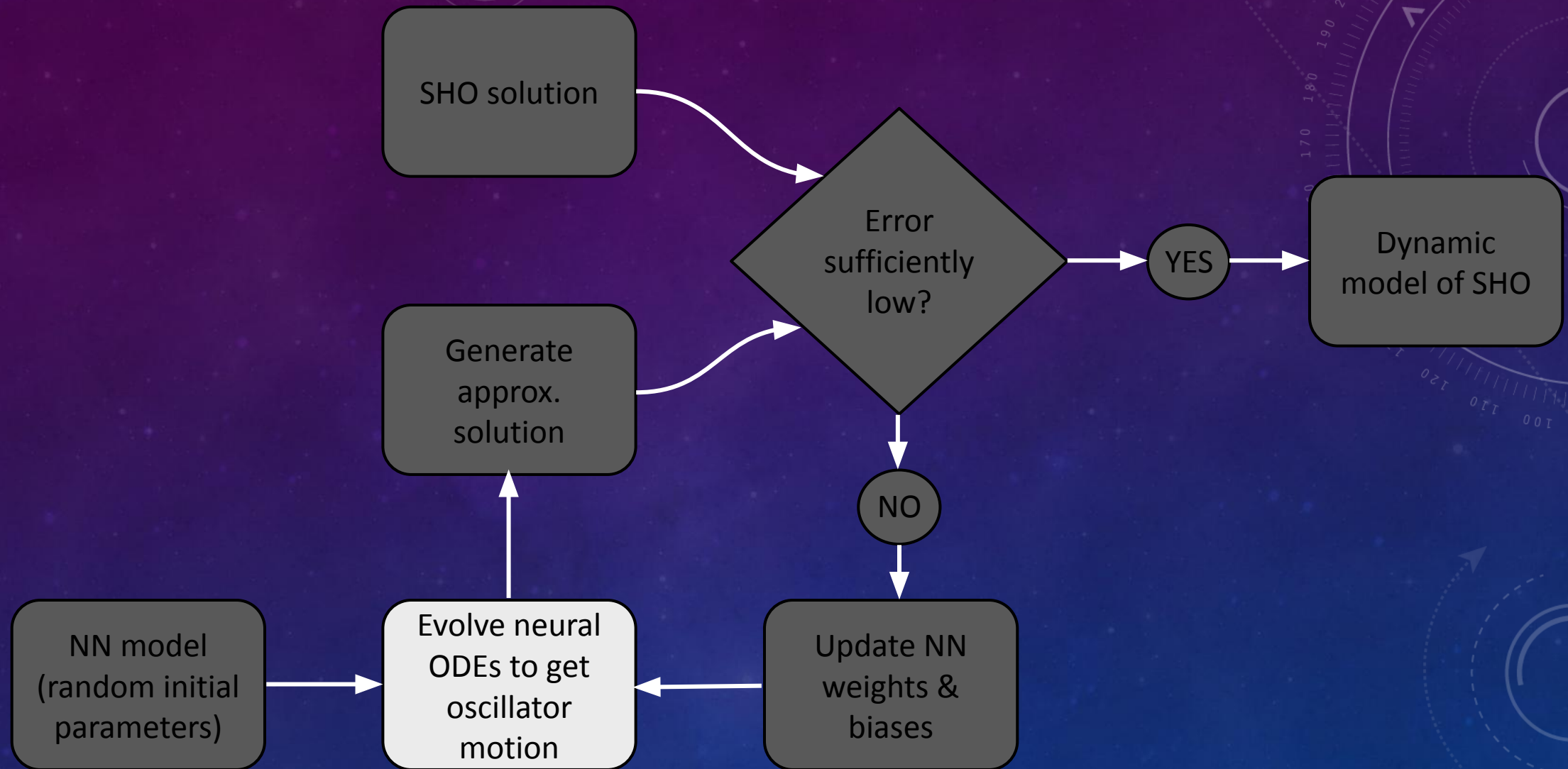
$$\lambda = [w_1, w_2, w_3, w_4, b_1, b_2, b_3]$$

Fig 11: The neural network parameters (Latex).

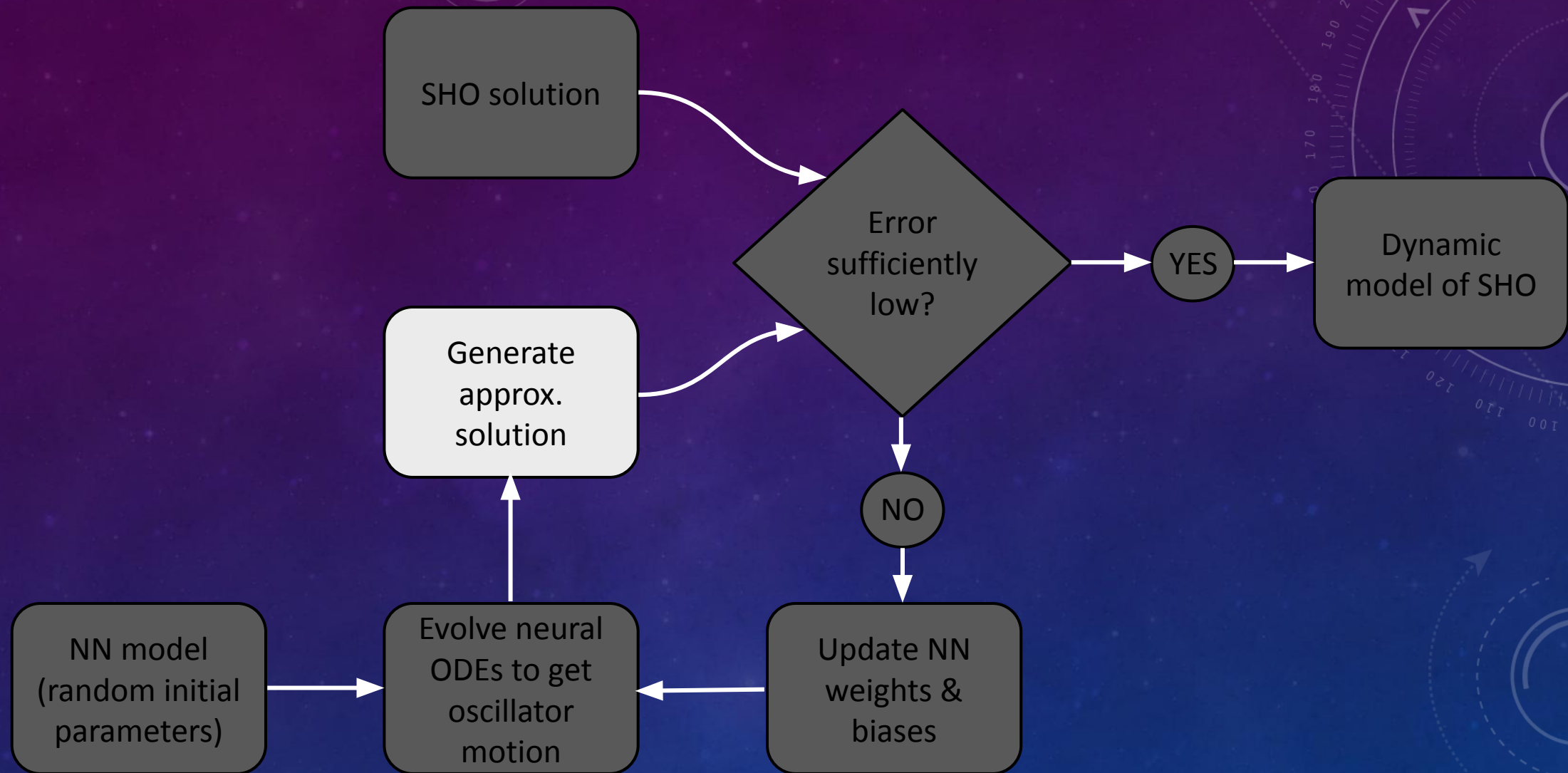
TRAINING PROCEDURE



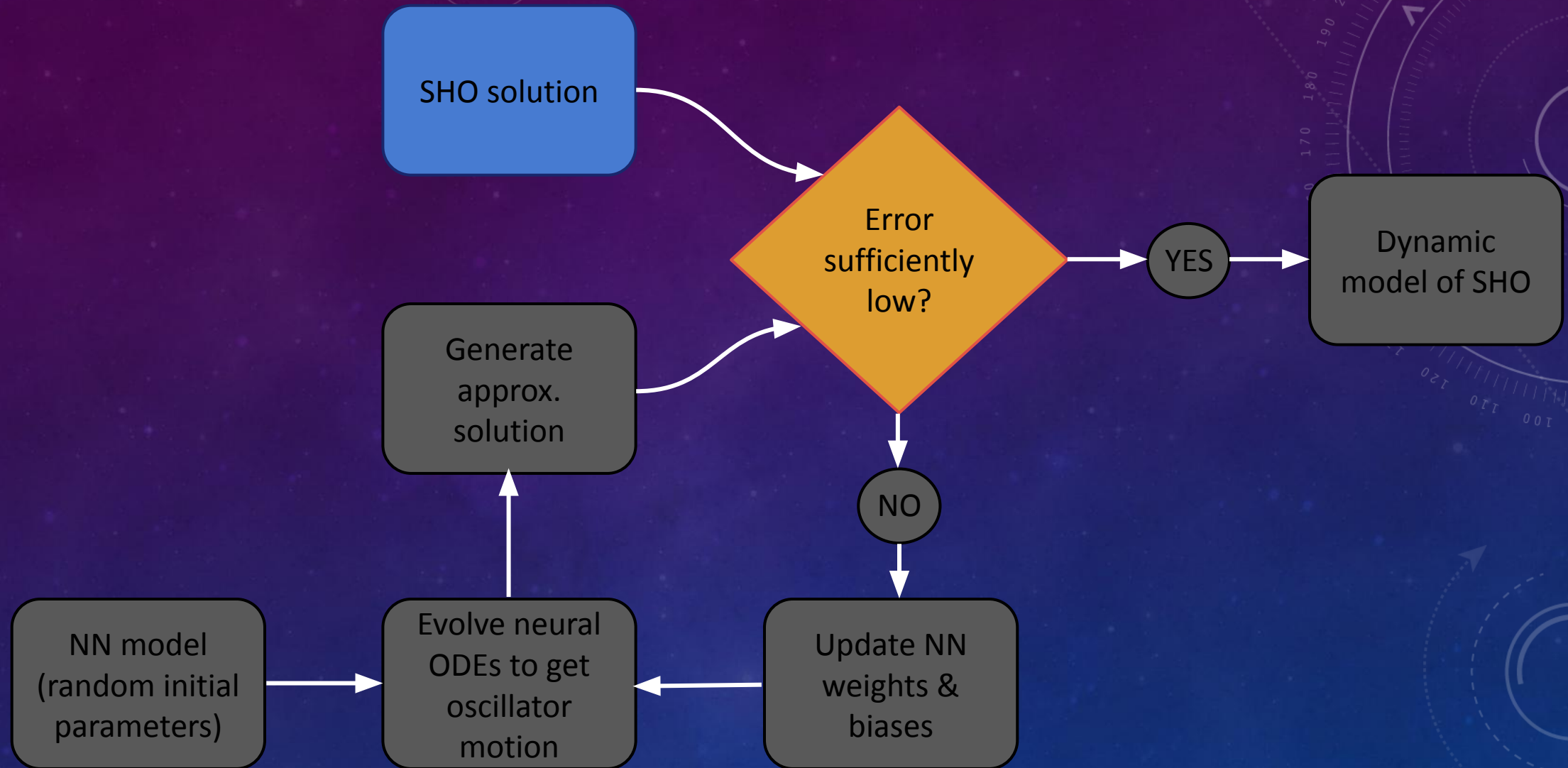
Step 1: Initialize network weights/biases



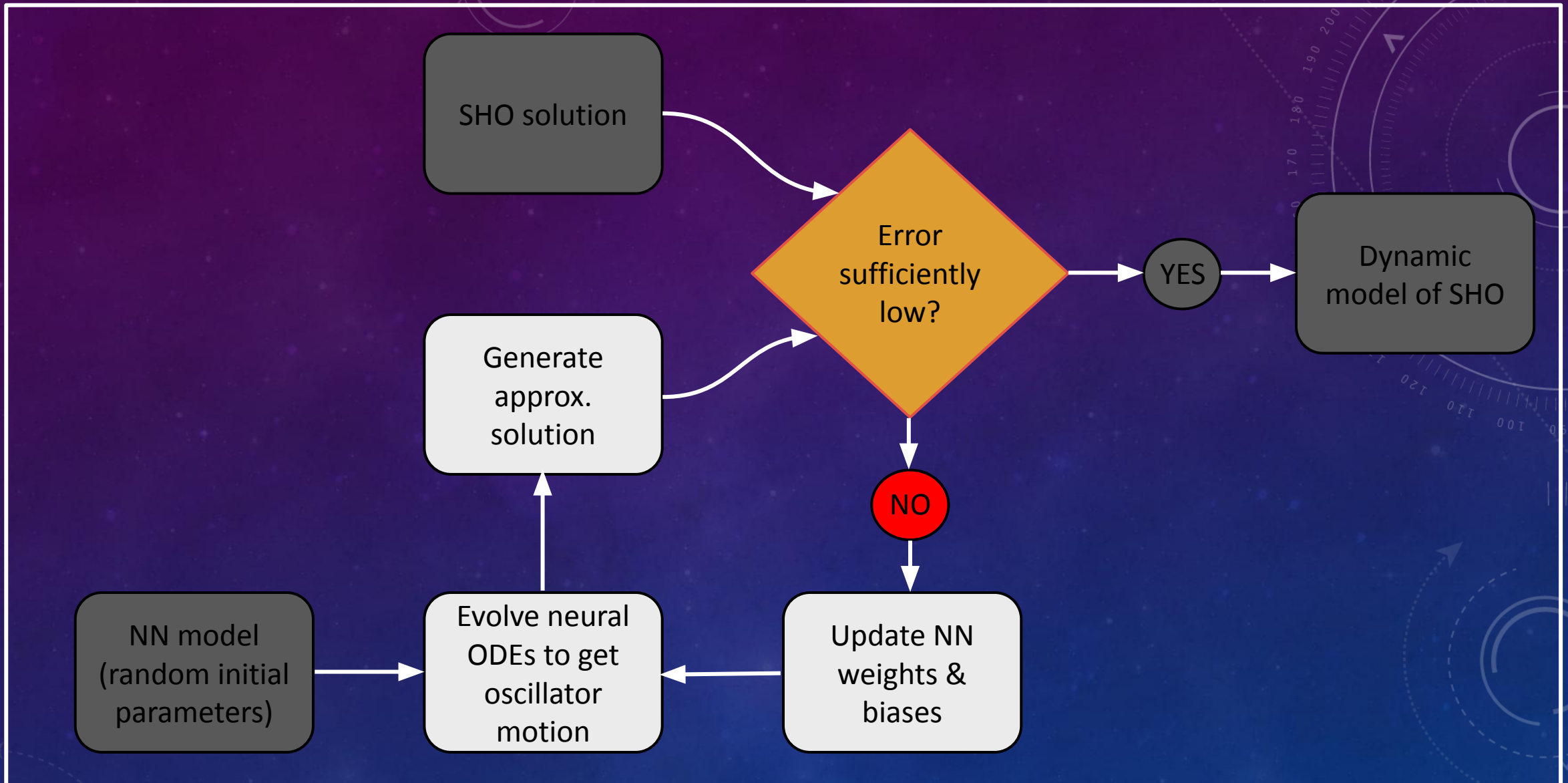
Step 2: Evolve system to find dx , dp



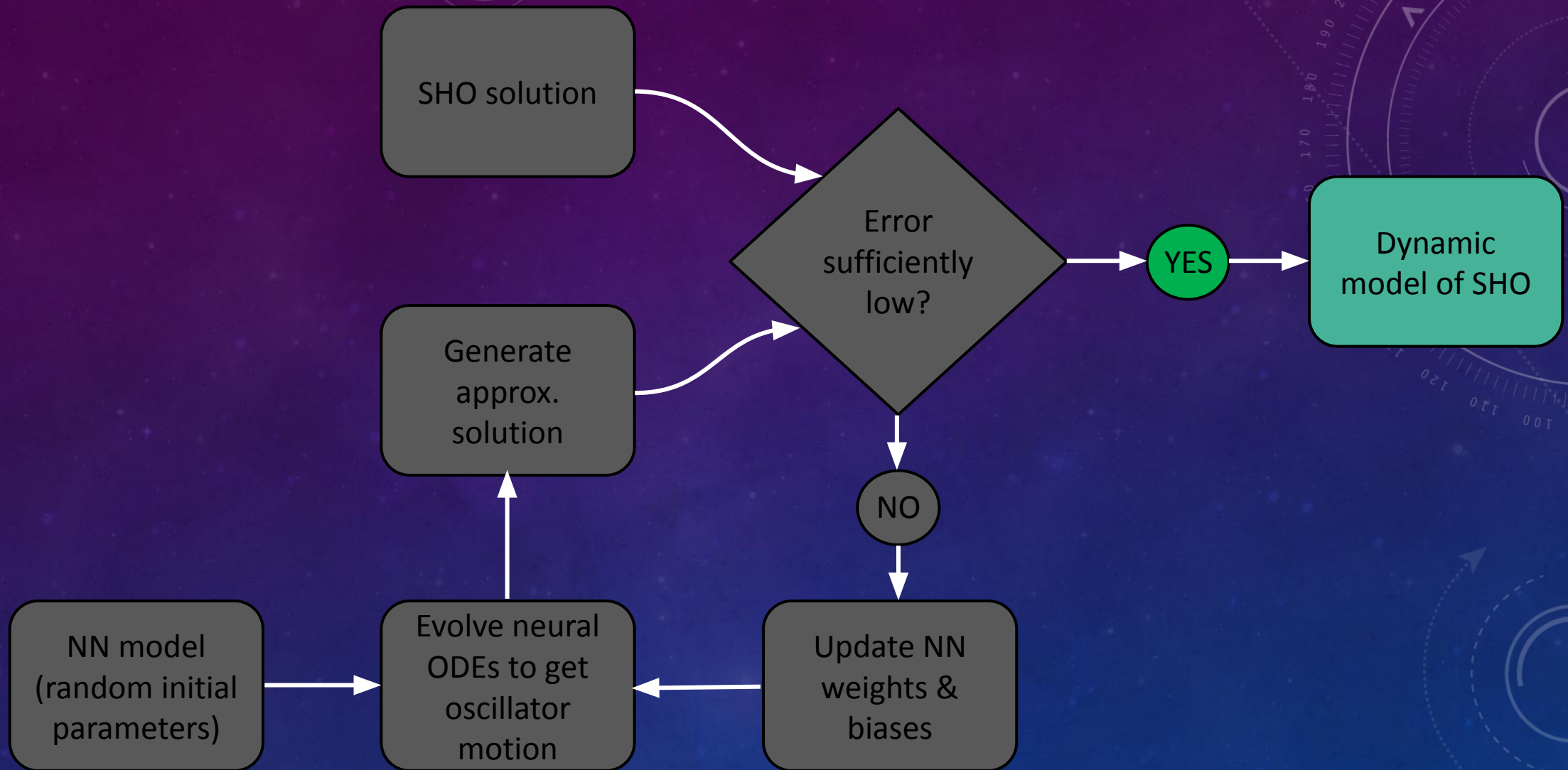
Step 3: Generate wave using NN



Step 4: Compare NN solution to true solution



Step 4: If error is not sufficiently low, return to step 2 and update weights/biases



Step 5: If error is sufficiently low, successful NN is found!

EXAMPLE

PRIOR TO TRAINING

Terminology:

- Datasize
 - Number of points
- Iterations
 - Number of times the neural network is trained
- Loss
 - Mean squared error
- Timespan
 - Length of time to be trained over
- NN parameters
 - Weights/biases of the NN

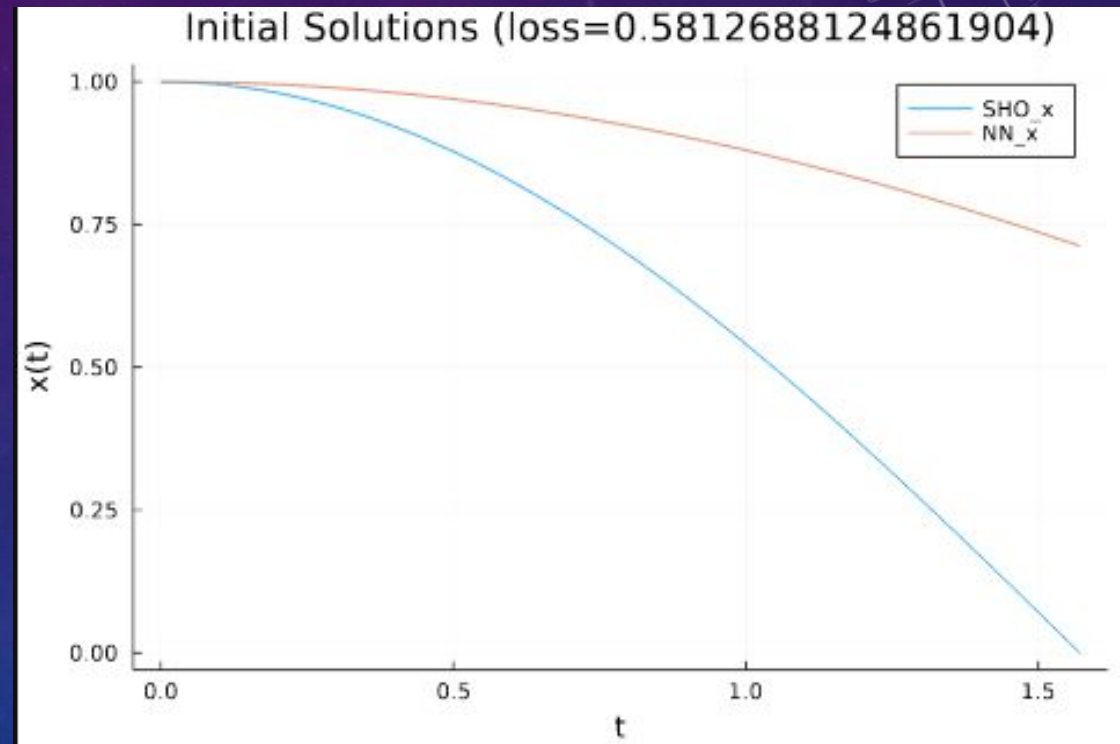


Fig 12: The output of the *untrained* NN compared to the SHO solution.

TRAINING

Sample Setup:

- Datasize
 - 300
- Iterations
 - 1200
- Timespan
 - $(0, \pi/2)$
- NN parameters
 - Random values between $(-1, 1)$

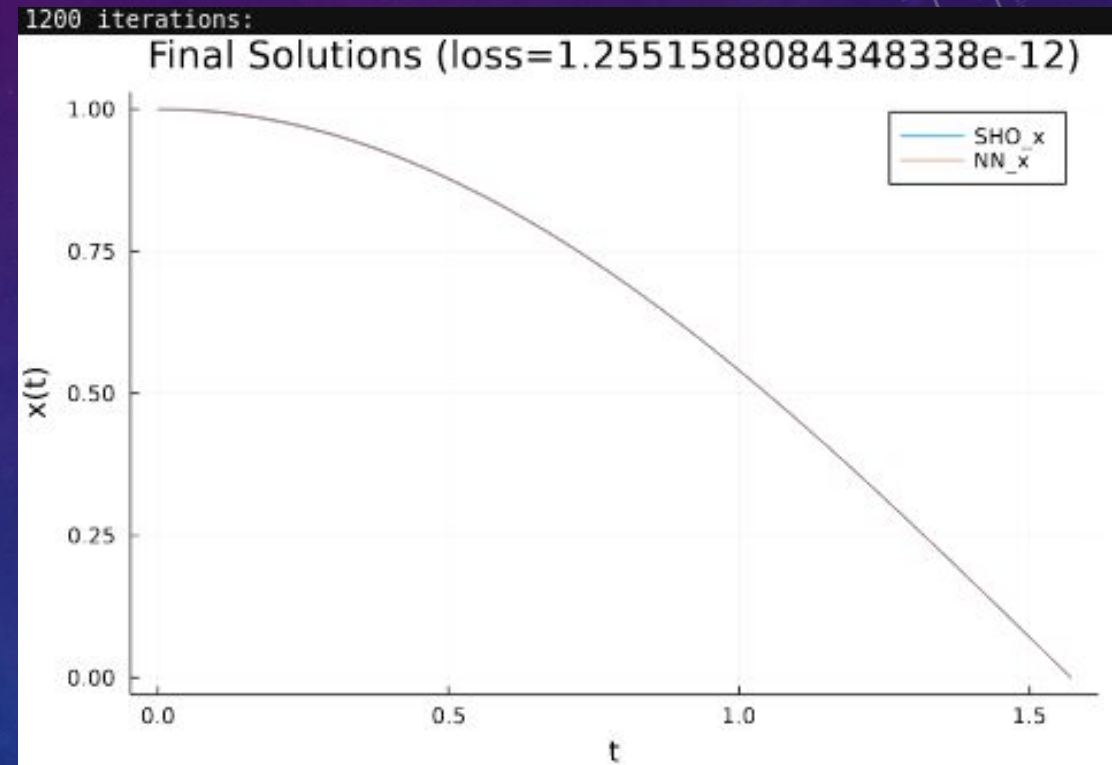
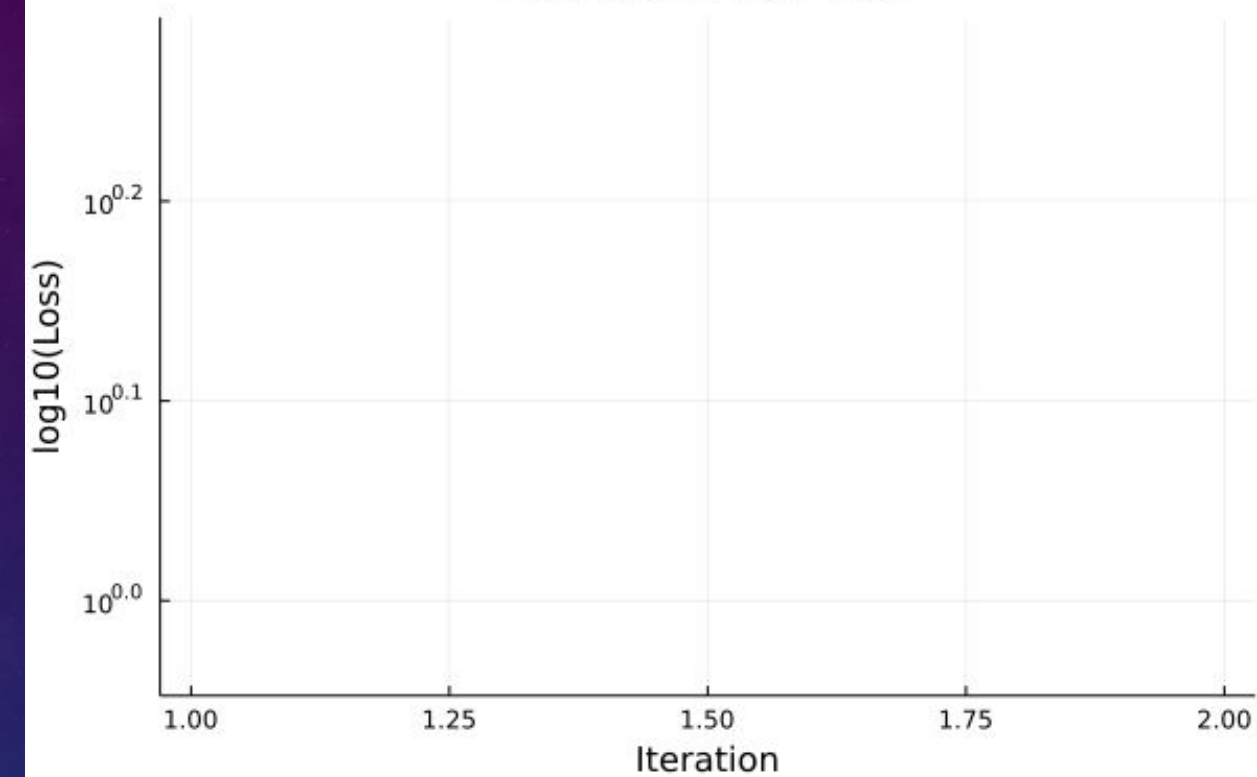


Fig 13: The output of the *trained* NN compared to the SHO solution.

TRAINING

Loss (iteration #1)



Solutions (iteration #1)

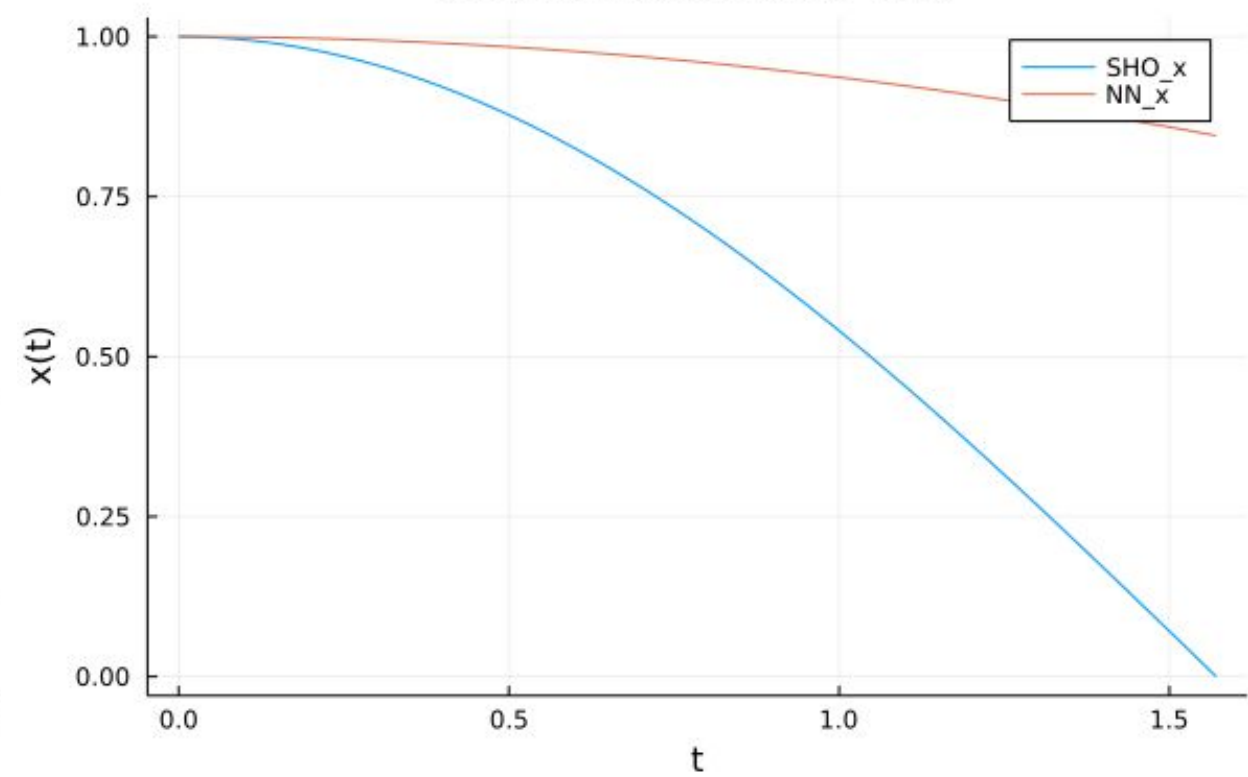


Fig 14: A real-time log10 plot of loss vs. the iteration (left). A real-time plot of the x solutions (right).

PROBLEMS TO AVOID

Iterations too small

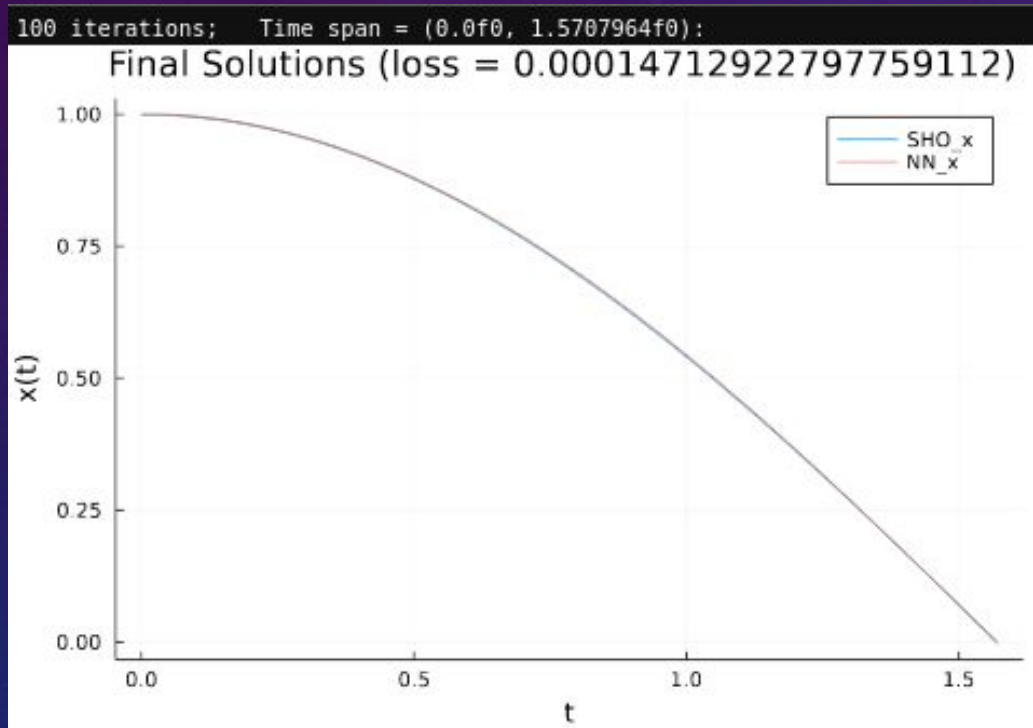


Fig 15: A result using too little max-iterations.

Time span too large

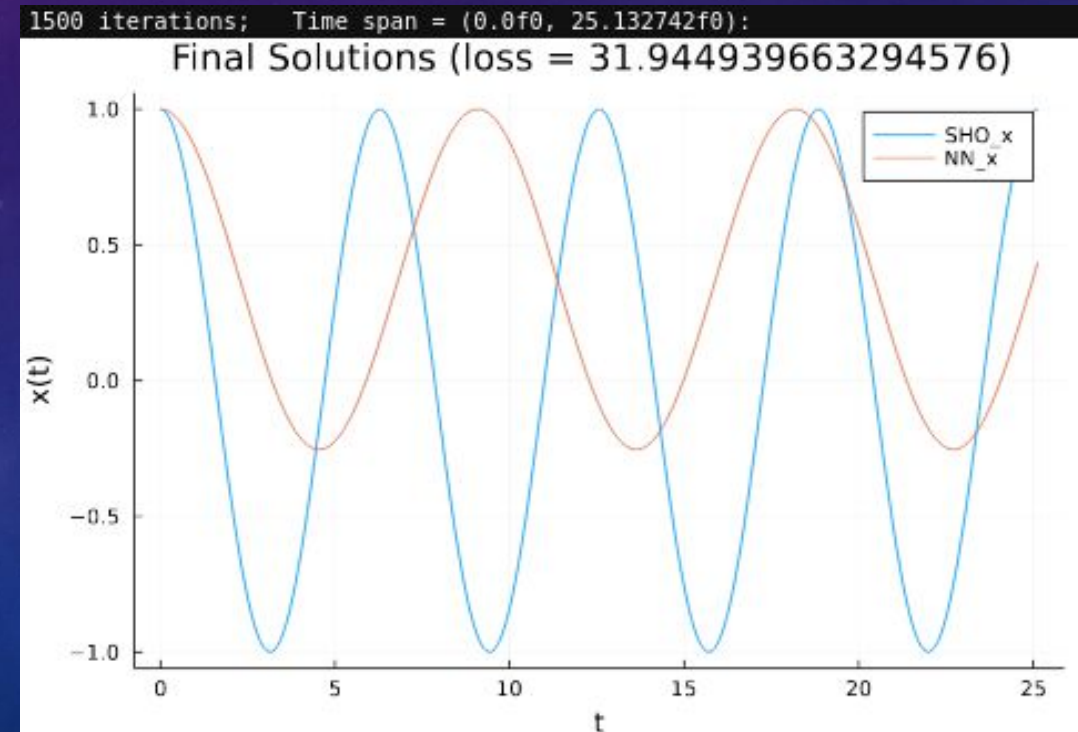


Fig 16: A result using too large of a time-span.

SUMMARY OF RESULTS

LARGELY DEPENDENT ON INITIAL NN PARAMETERS

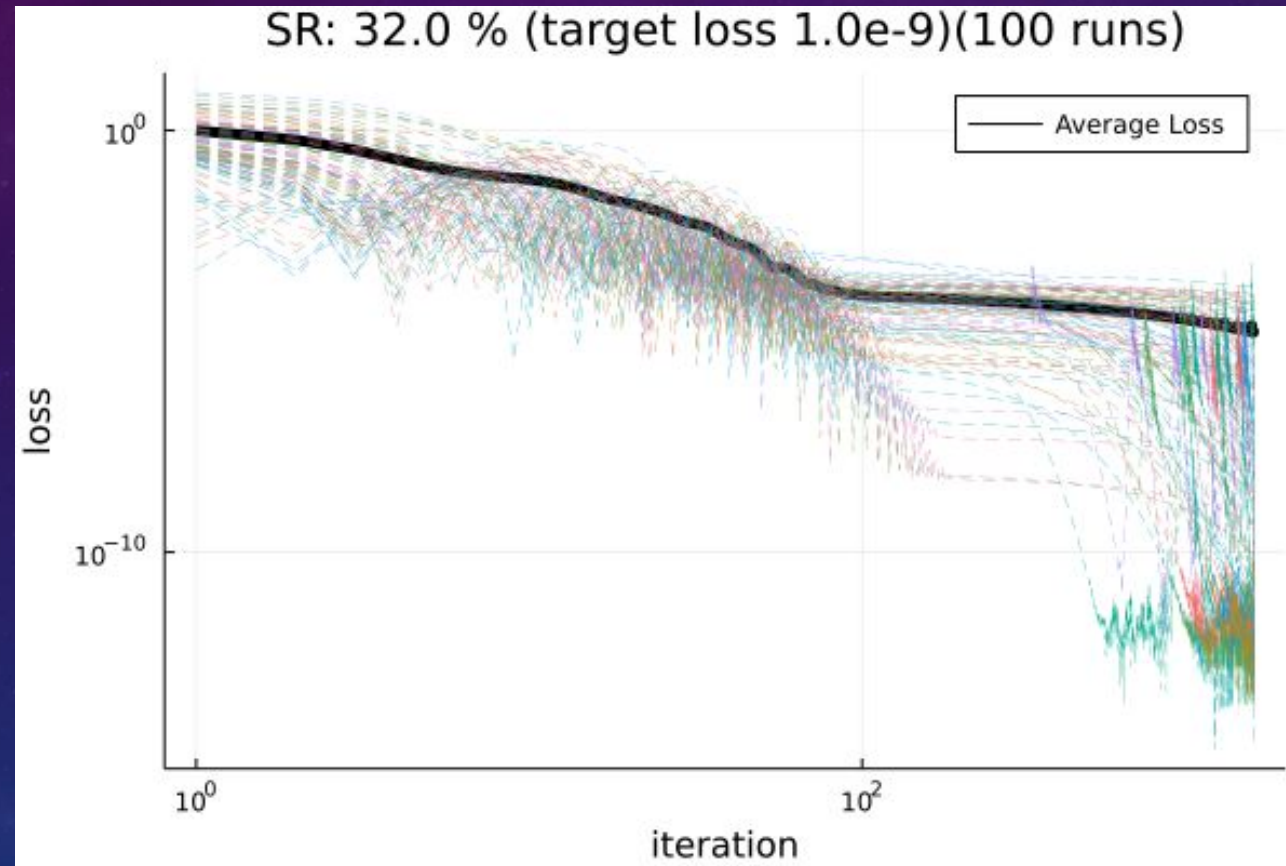


Fig 17: 100 runs with the same number of max-iterations, but different initial NN parameters.

IMPORTANCE OF CUMULATIVE LEARNING

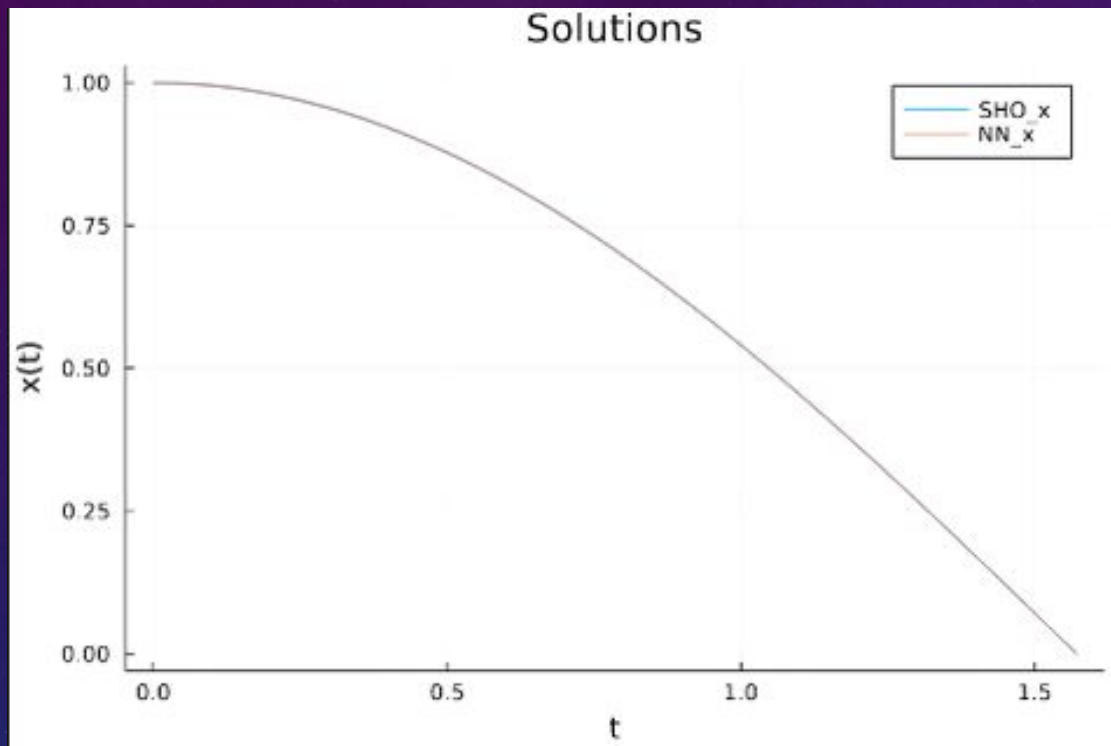


Fig 18: Cumulative learning example working up to 3 periods.

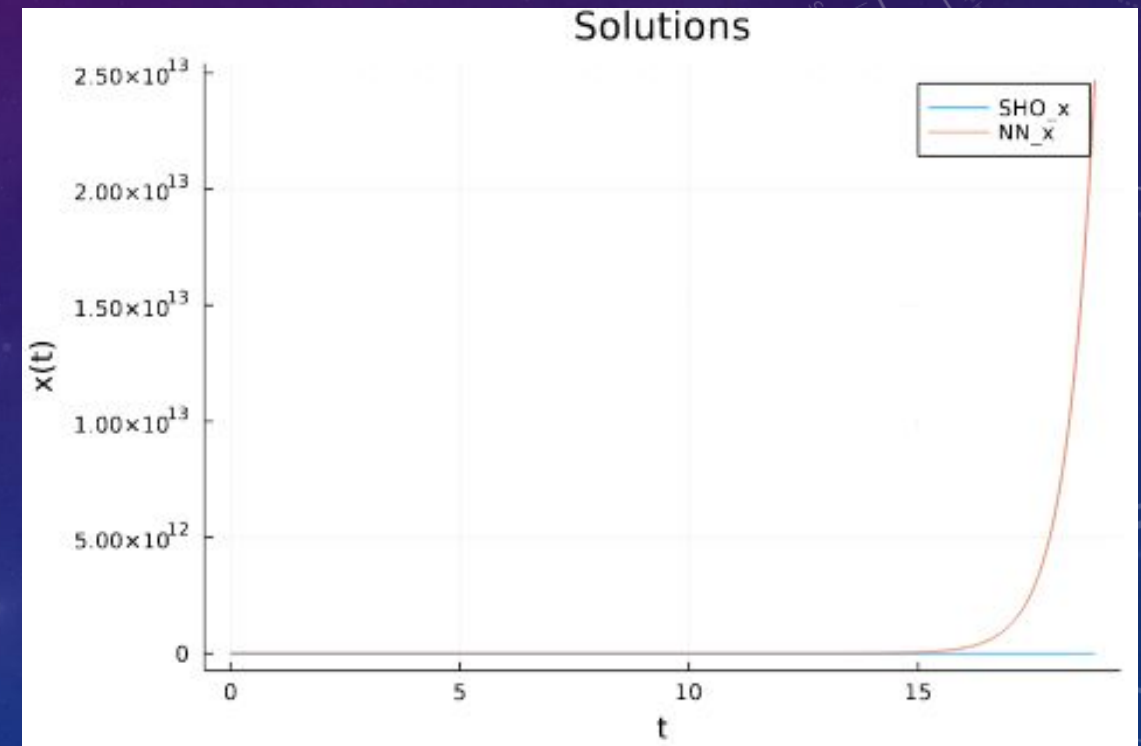


Fig 19: Different attempts to learn 3 periods directly

TAKEAWAY

- *NN with near perfect accuracy exists!*
- Can successfully learn ODE of SHO
 - Requires tweaking
 - Dependent on architecture, activation function, initial seed, learning rate
 - Once this is done, it works!

IMPORTANCE FOR BBHS

- First complicate the SHO
 - Dampening, variations in initial parameters, etc.
- Given time, tweaking can be streamlined
- Use the same process on GWs
 - Directly discover the ODEs (or Hamiltonian)
 - Ideally more efficient, faster

The background is a dark gray gradient. It features several abstract circular elements: concentric circles, dashed lines forming circles, and solid lines with arrows indicating a clockwise direction. Some of these circles have numerical markings around their perimeters, ranging from 0 to 210. The overall aesthetic is technical and modern.

THANK YOU!

QUESTIONS

WORK CITED

- Fig 3: <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.116.061102>
- Fig 5: <https://link.springer.com/content/pdf/10.12942/lrr-2014-2.pdf?pdf=button%20sticky>